

HASHING - 3

Start doing dry run,
start using pen and
paper while solving
DSA problems. Trust
me half of your
problems will be
solved.

Try this !!

Good
Evening
😊

Content for Today

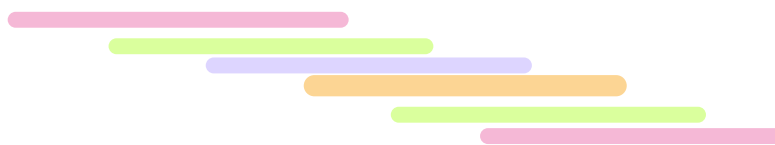
01. Distinct numbers in window
02. No. of distinct 2D points
03. Class object as key

Q1. Given N elements, calculate no. of distinct ele in every subarray of size k

arr[10] =

2	4	3	8	3	9	4	9	4	10
---	---	---	---	---	---	---	---	---	----

$k = 4$



<u>subarray</u>	No. of distinct ele
-----------------	---------------------

[0 - 3]	→ 4
---------	-----

[1 - 4]	→ 3
---------	-----

[2 - 5]	→ 3
---------	-----

[3 - 6]	→ 4
---------	-----

[4 - 7]	→ 3
---------	-----

[5 - 8]	→ 2
---------	-----

[6 - 9]	→ 3
---------	-----

Brute force Idea → For every subarray of size k , insert all the elements in hashset & get its size

$N=10$, $K=1$ # 10 subarrays

$N=10$, $K=2$ # 9 subarrays

$N=n$, $K=k$ # $n-k+1$ subarrays

For a single subarray of length $K \rightarrow O(K)$

TC = Total no. of subarrays * time for inserting one subarray

$$TC = (n-k+1) * k$$

Worst case

$$K=1 \Rightarrow (n-1+1) * 1 = O(n)$$

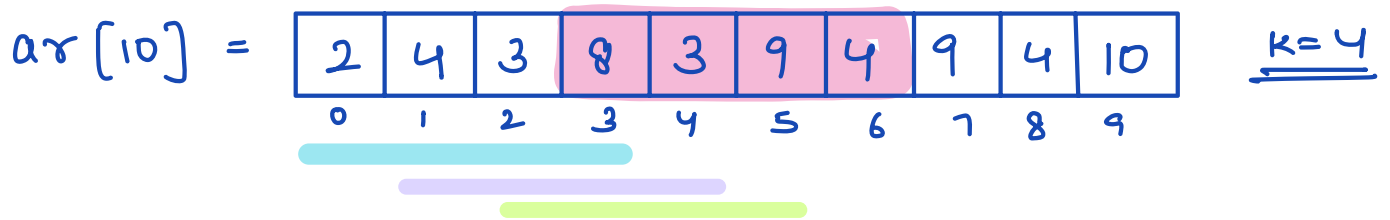
$$K=n \Rightarrow (n-n+1) * n = O(n)$$

$$K=n/2 \Rightarrow (n - \frac{n}{2} + 1) * (\frac{n}{2})$$

$$= (\frac{n}{2} + 1) (\frac{n}{2}) = \frac{n^2}{4} + \frac{n}{2} \Rightarrow O(n^2)$$

$$SC = O(K)$$

02. Sliding window + Hashset



Subarray

[0 - 3]		HS	Ans
		{ 2, 4, 3, 8 }	4
[1 - 4]	remove arr[0]	add arr[4] → { 4, 3, 8 }	3
[2 - 5]	arr[1]	arr[5] → { 3, 8, 9 }	3
[3 - 6]	arr[2]	arr[6] → { 8, 9, 4 }	3
			↓ Wrong

Problem → Frequency of each ele is not getting considered

HashMap

Idea 3 → Use sliding window + HashMap

Key → Distinct ele

Value → frequency

arr[10] =

2	4	3	8	3	9	4	9	4	10
0	1	2	3	4	5	6	7	8	9

 k=4

Subarr

HM

[0 - 3] → { <2,1> <4,1> <3,1> <8,1> } #4
remove add

[1 - 4] arr[0] arr[4] { ~~<2,0>~~ <4,1> <3,2> <8,1> } #3
↗ if freq == 0, remove

[2 - 5] arr[1] arr[5] { ~~<4,0>~~ <3,2> <8,1> <9,1> } #3

[3 - 6] arr[2] arr[6] { <3,1> <8,1> <9,1> <4,1> } #4

[4 - 7] arr[3] arr[7] { <3,1> ~~<8,0>~~ <9,2> <4,1> } #3

⋮
⋮
⋮

void distinct ele (int [] arr, int k)

HashMap<I, I> hm = new HashMap<>();

// Insert k elements

```
for (int i=0; i<K; i++) {
```

```

if (hm.containsKey(ar[i])){
    | hm.put(ar[i], hm.get(ar[i]) + 1);
    |
3
else
    hm.put(ar[i], 1);
3

```

```
print(hm.size());
```

```
// Work for all the other windows
```

$$S = 1, e = k$$

```
while ( e < n )
```

```
// get subarr from store  $\rightarrow$  remove arr[s-1]  
add arr[c]
```

```
hm.put (ar[s-1] , hm.get(ar[s-1])-1 );  
if (hm.get(ar[s-1]) == 0) hm.remove(ar[s-1]):
```

```

    if (hm.containsKey(ar[e])) {
        | hm.put(ar[e], hm.get(ar[e]) + 1);
    }

```

```

    }
    else {
        hm.put (ar[e], 1);
        s++;
    }
    e++;
}

```

print (hm.size());

s++, e++

$$TC = O(k + (n - k))$$

$$= O(n)$$

$$SC = O(k)$$

02. Given a 2D Array denoting points on a 2D plane.

Return total no. of distincts points in the array.

$A = \{ \overset{0}{\{5, 6\}} \overset{1}{\{2, 8\}} \overset{2}{\{2, 8\}}$
 $\{ -1, -1 \} \{ 2, -3 \} \{ 2, -3 \}$
 $\{ 2, 8 \} \{ 7, 7 \}$
 $\}$

HashSet

$\{ 5, 6 \}$
 $\downarrow \quad \downarrow$
 $x \quad y$

Distinct $\rightarrow \{ 5, 6 \} \{ 2, 8 \} \{ -1, -1 \} \{ 2, -3 \} \{ 7, 7 \}$

Ans $\Rightarrow 5$

01. Insert all points in HashMap <key, value>

HM < Integer, Integer > map = new HM <>();

(x)

(y)

Eg:- (2, 3) & (2, 8)

HM = { 2 → ~~3~~⁸ } Data can be
overwritten

02. Store complete point as a key

String → concatenate x & y & insert

HashSet <String> hs = new HashSet <>()

Edge case → (15, 2) & (1, 52)

"15" + "2"

"152"

"1" + "52"

"152"

HS will treat both of
them as same

Idea 3 → Concatenate the co-ordinates using a separator & add them as a string

(15, 2) & (1, 52)

"15" + "@" + "2"

"15@2"

"1" + "@" + "52"

"1@52"

HashSet is going to treat
both of them as different
strings

```
int 2Dpoints (int [][] ar)
```

```
    HashSet <String> hs = new HashSet<>();
```

```
    for (int i=0 ; i<ar.length; i++)
```

```
        x = ar[i][0]
```

```
        y = ar[i][1]
```

```
        String str = x + "@" + y ;
```

```
        hs.add(str);
```

```
    }
```

```
    return hs.size();
```

Tc : O(n)

Sc : O(n)

Object as a Key

How to put object as a key in hashmap?

Complete story of hashcode will get covered
in Implementation of HashMap

LinkedList

01. Every distinct key in the hashmap has
an integer value associated with it,

Hashcode

defined for every object in Java.

Pair sum = 5 $\rightarrow$

2 3

$$2 + 3 = 5$$

$$3 + 2 = 5$$

Pair diff = 1 $\rightarrow$

2 3

$$2 - 3 = -1$$

$$3 - 2 = 1$$